| | |
|---|---|
| **Trạng thái** | Đã xong |
| **Bắt đầu vào lúc** | Thứ Bảy, 17 tháng 1 2026, 12:51 PM |
| **Kết thúc lúc** | Thứ Bảy, 17 tháng 1 2026, 12:52 PM |
| **Thời gian thực hiện** | 43 giây |
| **Điểm** | 6,00/6,00 |
| **Điểm** | **10,00** trên 10,00 (**100**%) |

Câu hỏi **1**

Đúng

Đạt điểm 1,00 trên 1,00

Let p and q be two propositions. Consider the following two propositional logic:

A : $(\neg p \land (p \lor q)) \Rightarrow q$

B : $q \Rightarrow (\neg p \land (p \lor q))$

Which one of the following choices is correct?

- a. A is a tautology but B is not a tautology ✓
- b. Neither A nor B is a tautology
- c. Both A and B are tautologies.
- d. A is not a tautology but B is a tautology

Your answer is correct.

The correct answer is:
A is a tautology but B is not a tautology

Câu hỏi **2**

Đúng

Đạt điểm 1,00 trên 1,00

You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A, or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country. Express your answer in terms of

e: "You are eligible to be President of the U.S.A.,"

a: "You are at least 35 years old,"

b: "You were born in the U.S.A,"

p: "At the time of your birth, both of your parents were citizens,"

r: "You have lived at least 14 years in the U.S.A."

- a. $e \Rightarrow (a \wedge (b \vee p) \wedge r)$ ✓
- b. $(a \wedge (b \vee p) \wedge r) \Rightarrow e$
- c. $e \Rightarrow (a \wedge b \wedge p \wedge r)$
- d. $e \Rightarrow (a \wedge b) \vee (p \wedge r)$

Your answer is correct.

The correct answer is: $e \Rightarrow (a \wedge (b \vee p) \wedge r)$

Câu hỏi 3

Đúng

Đạt điểm 1,00 trên 1,00

Formalize the following statement using predicate logic: "At most one Democratic candidate can win the election.".

Given the following dictionary:

Cx : x is a candidate; Rx : x is Democratic; Wx : x can win the election.

- a. $\exists x((Cx \land Rx) \Rightarrow W x)$
- b. $\forall x((Cx \land Rx) \Rightarrow W x)$
- c. $\exists x((Cx \land Rx) \land \forall y(Cy \land Ry) \Rightarrow x = y)$
- d. $\forall x \forall y((((Rx \land Cx) \land W x) \land ((Ry \land Cy) \land W y)) \Rightarrow x = y)$ ✓

Your answer is correct.

The correct answer is: $\forall x \forall y((((Rx \land Cx) \land W x) \land ((Ry \land Cy) \land W y)) \Rightarrow x = y)$

Câu hỏi 4

Đúng

Đạt điểm 1,00 trên 1,00

Let Sx : x is a student; Cx : x is a computer; Oxy : x owns y
Express the following statement in a simple English sentence:

$\exists x(Sx \land \exists y \exists z(y \sim= z \land ((Cy \land Oxy) \land (Cz \land Oxz))))$

- a. Some students own more than one computers. ✓
- b. All of the students do not have a computer.
- c. Some students own exactly two computers.
- d. Some students own only one computer.

Your answer is correct.

The correct answer is: Some students own more than one computers.

Câu hỏi **5**

Đúng

Đạt điểm 1,00 trên 1,00

Which one of the following well formed formula is a tautology?

- a. $\forall x \forall y R(x, y) \Rightarrow \forall y \forall x R(y, x)$
- b. $[\forall x \exists y (P(x, y) \Rightarrow R(x, y))] \Longleftrightarrow [\forall x \exists y (\neg P(x, y) \lor R(x, y))]$
- c. $(\forall x [\exists y R(x, y) \Rightarrow S(x, y)]) \Rightarrow \forall x \exists y S(x, y)$
- d. $\forall x \exists y R(x, y) \Longleftrightarrow \exists y \forall x R(x, y)$

Your answer is correct.

The correct answer is: $[\forall x \exists y (P(x, y) \Rightarrow R(x, y))] \Longleftrightarrow [\forall x \exists y (\neg P(x, y) \lor R(x, y))]$

Câu hỏi **6**

Đúng

Đạt điểm 1,00 trên 1,00

The following proof of the theorem is correct or not ?

Theorem: For every positive integer n, if x and y are positive integers with max(x, y) = n, then x = y.

Proof :

1. Basis Step: Suppose that n = 1. If max(x, y) = 1 and x and y are positive integers, we have x = 1 and y = 1.
2. Inductive Step: Let k be a positive integer. Assume that whenever max(x, y) = k and x and y are positive integers, then x = y. Now let max(x, y) = k + 1, where x and y are positive integers. Then max(x − 1, y − 1) = k, so by the inductive hypothesis, x − 1 = y − 1. It follows that x = y, completing the inductive step.

- a. Incorrect at Basis step
- b. Both steps are incorrect
- c. Another solution. ✓ Incorrect at Inductive step
- d. Correct

Your answer is correct.

The correct answer is: Another solution.